Vietnam Math Modeling Competition 2026

SECOND ROUND – PART B

23/06/2026

ORGANIZER

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B1.

a)

If a strategy has m stints

(c1, n1), (c2, n2), (c3, n3), …, (c_m, n_m) with n1 + n2 + … + n_m = N

We have the total stint time:

$$S_c(n;s) = \sum_{k=1}^{n} \left[B_c + wF(s+k) + g_c(k)\big(1+\beta F(s+k)\big)\right]$$

$$= nB_c + \sum_{k=1}^{n} g_c(k) + w\sum_{k=1}^{n} F(s+k) + \beta \sum_{k=1}^{n} g_c(k)F(s+k)$$

The total direct fuel part is

$$\sum_{\ell=1}^{N} wF(\ell) = w\sum_{\ell=1}^{N}(N-\ell+1) = w\big(N+(N-1)+\cdots+1\big) = w\frac{N(N+1)}{2}$$

This is a constant.

So the effect of fuel on the weight of the car does not change the optimal strategy if the other parameters stay fixed. In contrast, the term β *g_c(k)F(l) makes the stint order meaningful.
If two stints i,j have lengths n_i,n_j and total degradation U_i,U_j, the cost difference between the order i then j and the order j then i is

$$\beta * n_i * n_j * (U_i/n_i - U_j/n_j).$$

Therefore, the stint with the smaller average degradation should be placed in the phase when the car is heavier.
Hence, increasing w shifts the time of every strategy by the same amount, so w does not change the optimal strategy.

b)

First, we compute

If $n \le L_C$:

$$S_c(k,l) = \sum_{k=1}^{n}(B_c + wF(l) + g_c(k)(1+\beta F(l))$$

$$= nB_c + wn(N-\ell_0-\frac{n-3}{2}) + d_c\cdot\frac{n(n-1)}{2}$$

$$+ d_c\beta((N-\ell_0)\frac{n(n-1)}{2} - \frac{n(n-1)(n-2)}{3})$$

By the lemma from problem A2, we consider a_c=(S_c+P)/n
Then:

$$\begin{aligned} a_c(n) &= S_c(k,l)/n \\ &= B_c + w(N - \ell_0 - \frac{n-3}{2}) + d_c \cdot \frac{n-1}{2} \\ &\quad + d_c\beta((N-\ell_0)\frac{n-1}{2} - \frac{(n-1)(n-2)}{3}) \end{aligned}$$

Setting a_c'(n)=0 to find the minimum of the function, differentiating in n gives:

$$n = \frac{w + d_c + d_c\beta(N - \ell_0 + 2)}{2}$$

Since w, d_c, β, N are constants, for each l_0 there exists exactly one n.
For every l_0 from 1 to N we get N values of n, and choose n* among these n values so that

$$S_c(n*) = min\{S_c(n1), S_c(n2), \ldots, S_c(nN)\}$$

We then consider the cases of m from 1 to N. Since the model has the shape of a parabola with a second derivative greater than 0, when the time starts to decrease we stop considering and choose n*. Then n* will be the minimum of the equation
For each m, we consider m-1 pit stops at m-1 different laps of the race.
For example: with m=4, N=70, we can consider pit stops at laps (10, 20, 46); (1, 1, 68); …
Then, we choose the l_0 of each stint to be the start of the lap after the pit stop. For example, as above, we choose l_0 as 0, 10, 20, 46 for the 4 stints.
These stints are completely independent of each other.
Using the way n was found above, we find the optimal n for each position l_0 and m.

c) Applying the strategy presented above with the parameters

N=50, P =25, w=0.06, β=0.01.

We obtain the optimal solution:

M20 + S15 + S15, TB = 4219.92 s

B2.

Preliminary remarks:

For the same race, when only w is changed from 0.06 to 0.10 while keeping β = 0.01, the strategy does not change.

$$\Delta T = (0.10 - 0.06)\frac{N(N+1)}{2} = 0.04\frac{N(N+1)}{2},$$

However, when β is increased from 0.01 to 0.03, the strategy changes for many races. This is because the car is then heavier at the start of the race, so it is often switched to Hard earlier.

Results:

Figure 1: Results for all 3 cases. The strategies of B2(a) and B2(b) are the same and differ only in time (proved above)



Figure 2: Time results for all 3 cases

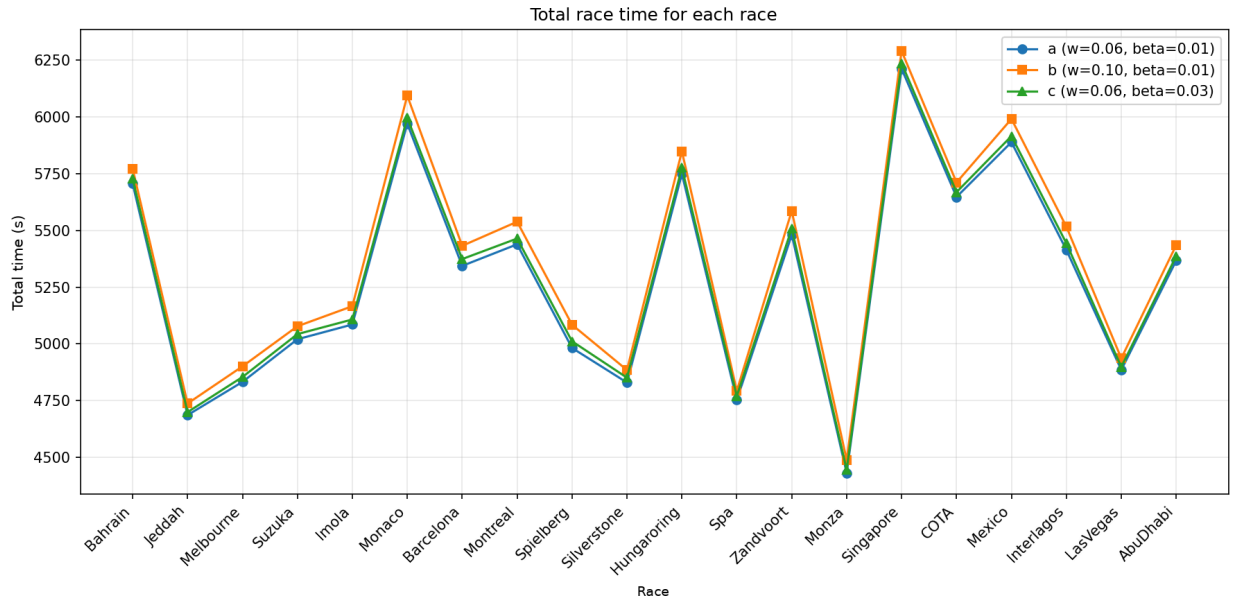


Table 1: Exact time results for all 3 cases

Track	a (w=0.06, beta=0.01)	b (w=0.10, beta=0.01)	c (w=0.06, beta=0.03)
Bahrain	5705.39	5771.51	5728.07
Jeddah	4686.59	4737.59	4699.75
Melbourne	4832.78	4901.22	4854.41
Suzuka	5020.59	5077.82	5043.22
Imola	5085.48	5166.12	5107.68
Monaco	5969.70	6092.94	5995.64
Barcelona	5343.52	5431.96	5373.67
Montreal	5438.56	5537.96	5463.98
Spielberg	4982.19	5084.43	5011.59
Silverstone	4829.94	4885.06	4851.39
Hungaroring	5748.61	5848.01	5775.55
Spa	4753.81	4793.41	4770.28
Zandvoort	5480.29	5585.41	5508.46
Monza	4431.91	4489.15	4445.22
Singapore	6211.83	6289.95	6236.13
COTA	5647.06	5710.90	5668.18
Mexico	5888.42	5990.66	5915.06
Interlagos	5413.94	5516.18	5443.22
LasVegas	4885.59	4936.59	4898.75
AbuDhabi	5366.78	5435.22	5386.74

Remarks:

When β is increased from 0.01 to 0.03, the total number of stints does not change, but the compound mix tends to shift toward more durable tires:

- Hard stints almost double: 18 to 35.

- Soft stints fall to one third: from 12 to 4.
- Medium stints decrease: 35 to 26.
- The number of races still using the Soft tire decreases from 7 to 3.

Of the 20 races, 14 races completely change their compound sequence, while the remaining 6 races (Jeddah, Monaco, Montreal, Monza, Las Vegas, Spa) only readjust how the laps are divided and keep the same compound sequence. Most of these 6 stable races already ran Hard-leaning strategies and almost never used Soft, so β only fine-tunes the pit points.

However, the time penalties caused by the two parameters differ significantly:

- Changing w (0.06 to 0.10): +77.96 seconds per race on average, largest at Monaco (+123.24 seconds, as N = 78 is the largest) and smallest at Spa (+39.60 seconds, N = 44). As this term is proportional to N(N+1)/2, the longer the race, the more is added.
- Changing β (0.01 to 0.03): on average only +22.70 seconds per race, ranging from 13.16 to 30.15 seconds.

B3.

Evaluation:

+ Strengths:

Strength	Explanation
Easy to understand	The two new parameters w and β have a clear physical meaning. A heavy car is slower and wears tires faster
Simple, easy to compute, light	The algorithm is relatively simple and does not require elaborate or advanced algorithms
Captures the stint order	Since F(l) decreases over time, the same stint placed at the start or at the end of the race can give different results

+ Weaknesses:

Weakness	Explanation
F(l) = N − l + 1 does not yet capture all the factors	Actual fuel use does not depend only on the number of remaining laps. Driving style, the Safety Car or the type of car can change the consumption rate
w does not change the strategy	This may be a feature of the problem statement, but it is still a major flaw that w does not affect the strategy
Does not consider external factors	Compared with our team's A6, this model does not consider the other external factors that we included.

Proposed improvement:

One improvement that still keeps the simplicity is to let the parameters depend on the tire compound:

Instead of using common w and β, we use

w_c and β_c

for each compound c ∈ {S, M ,H}

At the same time, replace the piecewise wear function with a smooth function:

$$g_c^{\text{new}}(k) = d_c(k-1) + A_c \ln\left(1 + e^{\lambda_c(k-L_c)}\right)$$

Where:

- d_c(k-1) is the initial uniform wear part
- A_c adjusts the severity of the phase in which the tire drops off quickly
- lambda_c adjusts the sharpness of the transition
- L_c is still the cliff where performance starts to drop sharply

In addition, we propose adding a cold-tire effect after a pit stop using a decaying function:

$$u_c e^{-\rho_c(k-1)}$$

The newly proposed full model is

$$t_c^{\text{new}}(k,\ell) = B_c + w_c F(\ell) + g_c^{\text{new}}(k)\left(1 + \beta_c F(\ell)\right) + u_c e^{-\rho_c(k-1)}$$

Improvements of the new model:

Improvement	Explanation
Use a separate w_c per tire	With a heavy car, each compound may be affected differently. The current fuel load can also affect the tire choice.
Use a separate β_c per tire	Different tires may be differently sensitive to the direct fuel at that moment. The new model handles this issue
Use a smooth g^new_c(k)	Avoids an abrupt change in tire performance at the cliff
Use u_c.e^−p_c(k−1)	This variable describes the new-tire effect after the pit

Acknowledgments.

Our team sincerely thanks the Organizing Committee and the Board of Judges for creating an academic competition in which students can approach applied mathematics, build a scientific report and apply mathematical models to a real-world problem. Throughout the process, the team practiced turning data and assumptions into a quantitative model, evaluating errors and

linking mathematical results with strategic decisions. The feedback of the teachers is a valuable basis for the team to continue improving its modeling thinking.